



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025-2026
HALF YEARLY EXAMINATION

CLASS: X
SUBJECT: PHYSICS [SET A]

FULL MARKS: 80
TIME: 2 HOURS

*Candidates are allowed additional 15 minutes for only reading the paper.
They must NOT start writing during this time.
The intended marks for questions or parts of questions are given in brackets []
Section A is compulsory. Attempt any four questions from Section B
This paper consists of eight printed pages.*

SECTION A (40 Marks)
(Attempt all questions from this Section.)

Question 1

[15 × 1 = 15]

Choose the correct answers to the questions from the given options:

- (i) In which of the following situations is the moment of force zero?
 - (a) A force is applied perpendicular to a lever at its free end.
 - (b) A force is applied at a distance from the pivot.
 - (c) A force is applied at the point of rotation.
 - (d) A force is applied tangentially to a circular wheel.
- (ii) A toy car moves at a constant speed along a circular track. When the toy car suddenly stops, it tends to move forward in a straight line rather than following the circular path. Which of the following best explains this behaviour?
 - (a) The car's engine forces it to move forward.
 - (b) Gravity pulls the car straight ahead when it stops.
 - (c) The car is pushed by an external force in a straight line.
 - (d) The inertia of the car causes it to continue moving in a straight line.
- (iii) The motion of a shock absorber in vehicles is an example of:
 - (a) free vibration (b) damped vibration (c) forced vibration (d) resonant vibration
- (iv) While loading a truck, a worker places heavy boxes on the upper shelves and lighter items at the bottom. As the truck begins to move, it becomes unstable and tilts during turns. What was the likely mistake made during loading?
 - (a) The weight of the truck became too low.
 - (b) The centre of gravity was lowered too much.
 - (c) The centre of gravity was raised.
 - (d) The load increased friction between the tyres and the road.
- (v) The electromagnetic wave with highest energy amongst the following is
 - (a) UV ray (b) Infrared ray (c) Gamma ray (d) X-ray

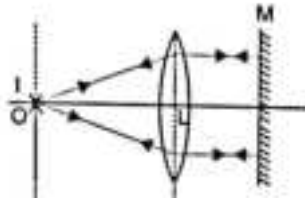
- (vi) A person climbs a flight of stairs slowly, while another person runs up the same stairs quickly. Which of the following is true about the work done by both individuals.
- The person who runs does more work because he uses more energy.
 - The person who climbs slowly does more work because it takes longer.
 - Both do the same amount of work because work depends only on vertical height and weight.
 - Work done is zero in both cases because they return to rest after reaching the top.

(vii) Assertion (A) : A convex lens can be used as diverging lens.

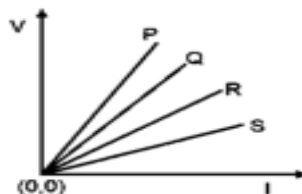
Reason(R) : The lens is placed in a medium with a lower index of refraction than that of the lens, hence the focal length becomes negative.

- Both A and R are true, and R is the correct explanation of A.
- Both A and R are true, and R is not the correct explanation of A.
- A is true, but R is false.
- A is false, but R is true.

(viii) Choose the correct option with respect to the following diagram:



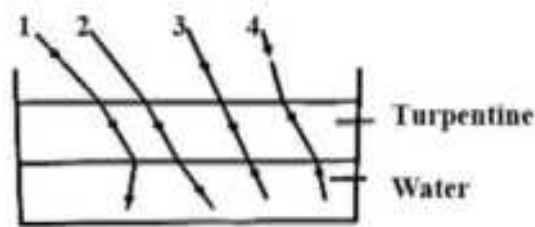
- If the distance of the plane mirror from the lens is increased, position of the image will change.
 - If the distance of the object from the lens is increased, object and image will no longer coincide.
 - No change will be observed if the lens is replaced by a concave lens.
 - No change will be observed if the plane mirror is replaced by a glass block.
- (ix) A worker pushes a heavy box across a floor at constant speed. Which statement correctly explains the work done and energy changes involved?
- The worker does positive work to move the box and energy is used to overcome friction.
 - The worker does no work since the speed is constant, hence no energy is used.
 - The worker does negative work because friction opposes motion.
 - The worker gains energy from pushing the box.
- (x) The graph of voltage vs current for four different materials is shown below:



Which of these four materials would be used for making the coil of a toaster?

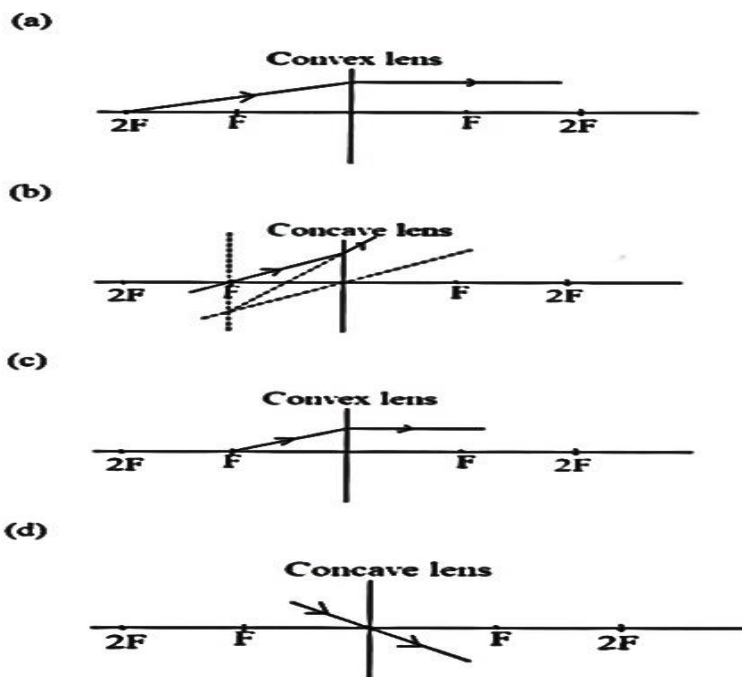
- Q
- S
- P
- R

- (xi) The optical density of turpentine is higher than that of water. The figure given below shows a layer of turpentine floating over water taken in a container. For which one of the four rays incident on turpentine in the path shown in the figure is correct?



- (a) 1 (b) 2 (c) 3 (d) 4

- (xii) Find the incorrect diagram:



- (xiii) White light passes from a glass prism into air. It is observed that blue light undergoes total internal reflection at a smaller angle of incidence compared to red light. Why does this happen?
- (a) Blue light has a higher refractive index in glass than red light, resulting in a smaller critical angle.
 - (b) Red light has a higher frequency and therefore bends more in glass.
 - (c) The critical angle is the same for all colors in glass because it depends only on the medium.
 - (d) Red light travels faster in air, so it escapes the prism more easily.
- (xiv) An engineer designs a building in a region prone to earthquakes. To prevent damage from resonance, what design consideration should be prioritized?
- (a) Making the building as tall as possible to absorb vibrations.
 - (b) Using materials that amplify oscillations at natural frequency.

- (c) Designing the building without any damping mechanism.
 (d) Ensuring the building's natural frequency does not match the frequency of seismic waves.
- (xv) A student notices that when two identical bulbs are connected in series to a battery, both bulbs glow dimly, but when connected in parallel to the same battery, both of them glow brightly. Which of the following explanations best describes this observation?
- (a) In series, the voltage divides between the bulbs reducing brightness; in parallel, each bulb gets full voltage.
 (b) In series, the current increases causing dim light but in parallel, current decreases causing bright light.
 (c) In series, the resistance decreases; in parallel, resistance increases causing brightness.
 (d) Both bulbs behave the same in series and parallel circuits.

Question 2

- (i) Complete the following by choosing the correct answers from the bracket: [6]
- (a) While entering from medium A to medium B, light slows down then ____ [$\angle i > \angle r$ / $\angle i < \angle r$ / $\angle i = \angle r$]
 (b) The characteristic of sound which enables to differentiate between two sounds of different intensity is ____ [*quality/pitch/loudness*].
 (c) The Potential difference between a live wire and a neutral wire is ____ [*0V/ 110 V/ 220V*]
 (d) The ratio of velocities of light of wavelength 400 nm and 800 nm in vacuum is ----- [*1:1/ 1:2/ 2:1*]
 (e) If a block and tackle system has three pulleys in the movable block, then its velocity ratio will be ____ [*6 / 7/ 6 or 7*].
 (f) A simple pendulum oscillating in air is an example of ____ [*natural vibration/damped vibration/forced vibration*].
- (ii) Match the examples of energy conversion in Column A to the type of energy conversion in Column B. [2]

Examples of energy conversion Column A	Type of energy conversion Column B
(a) An electric cell in use	• Chemical to heat
(b) Respiration	• Heat to mechanical
	• Chemical to electrical

- (iii) An object is seen first in red light and then in violet light through a simple microscope.
 (a) In which case is the magnification larger?
 (b) Justify your answer. [2]

Question 3

- (i) (a) The layered lens as shown is made of two different transparent materials. If a point object is placed on its axis, how many images does it form?



- (b) Give reason to your answer. [2]

- (ii) The critical angle of glass-air interface is θ_c .
 (a) Will the critical angle for glass-water interface be greater than or less than θ_c ?
 (b) Justify your answer. [2]
- (iii) Two strings of diameter 1 cm and 1.5 cm are set into vibration.
 (a) Which of them will produce sound of higher pitch?
 (b) Give reason to your answer. [2]
- (iv) (a) Why does increasing the number of pulleys in a system not always result in a proportional increase in efficiency?
 (b) Mention one factor that limit the efficiency of pulley systems in real-life applications. [2]
- (v) A coin placed at the bottom of a beaker appears to be raised by 8.0 cm. If the refractive index of water is $\frac{4}{3}$, find the depth of the water in the beaker. [2]
- (vi) The wavelength of waves produced on the surface of water is 20 cm. If the wave velocity is 24 ms^{-1} , calculate:
 (a) the number of waves produced in one second and
 (b) the time required to produce a wave. [2]
- (vii) A lever is a simple machine consisting of a rigid bar that rotates around a fixed point called the fulcrum. It is used to make work easier by either increasing force or increasing speed of movement, depending on its configuration and set up. A force multiplier is used to lift a load by applying a small effort over a longer distance.
 (a) Does a cricket bat function as a speed multiplier or a force multiplier?
 (b) Explain how any type of lever follows the principle of conservation of energy.
 (c) Give an example of a lever that changes the direction of the applied force. [3]

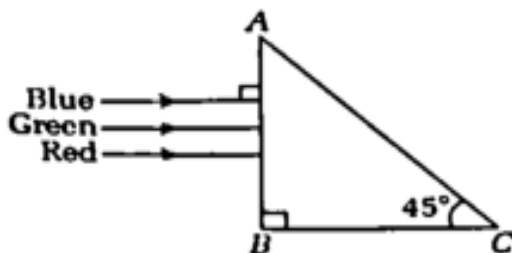
SECTION B

(Attempt any four questions)

Question 4

[3+3+4]

- (i) Given that the critical angle for glass-air interface for green light is 45° , complete the diagram to show the refracted and emergent rays. The emergent rays should be labelled according to their color.

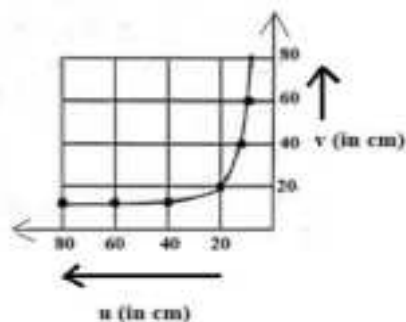


- (ii) (a) During heavy pollution, why does the sky appear more reddish or brownish compared to a clear day? [Relate your answer to scattering.]
 (b) When does a ray passing through a prism deviate away from the base of the prism?
 (c) Give one advantage of a total reflecting prism over a plane mirror.
- (iii) A virtual, diminished image is formed when an object is placed between the optical centre and the principal focus of a lens.
 (a) Name the type of lens which forms the above image.
 (b) Draw a ray diagram to show the formation of the image with the above stated characteristics.

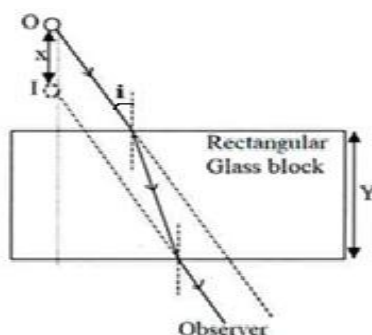
Question 5

[3+3+4]

- (i) A lens creates an image of an object. The distance of the object to the lens is u cm and the distance of the image is v cm. The given graph shows the variation of v with u .
- (a) Find the nature of the lens.
- (b) Determine the focal length of the lens.



(ii)



The above diagram shows that an observer sees the image of an object O at I.

- (a) Name the phenomenon responsible for seeing the image at a different position.
- (b) State the effect on X when:
1. Y increases.
 2. $\angle i$ decreases.
- (iii) (a) Name the radiations:
1. used as one of the diagnostic means in medical field.
 2. used in water purifiers.
 3. which is absorbed by the ozone layer in the atmosphere.
- (b) What is the frequency range of visible series of electromagnetic spectrum in kHz?

Question 6

[3+3+4]

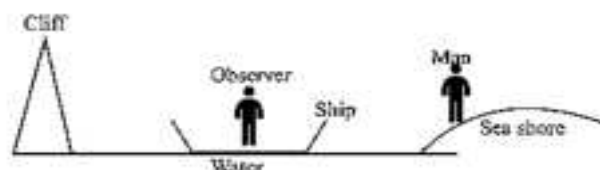
- (i) (a) Though same current flows through the electric line wires and the bulb filament, yet only the filament glows. Why?
- (b) Name the material of the wire used in the fuse.
- (c) Mention two key properties of the material that make it suitable for the intended use.
- (ii) A uniform meter scale is balanced at the 40 cm mark when a 100 g weight is suspended at the 10 cm mark.
- (a) What is the mass of the meter scale?
- (b) At what point would the scale balance if the 100 g weight is removed?
- (iii) A simple pendulum bob of mass 0.5 kg is released from a height of 0.4 m above its lowest position. Assume that there is no air resistance and take $g = 9.8 \text{ ms}^{-2}$.
- (a) Calculate the potential energy of the bob at its highest point.

- (b) What is the total mechanical energy of the pendulum during its swing?
 (c) Using mechanical energy conservation, find the speed of the bob at the lowest point.

Question 7

[3+3+4]

- (i) (a) When a vibrating tuning fork is held close to ear, one hears a faint hum. The same vibrating tuning fork is held such that its stem is in contact with the table surface, then one hears a loud sound. Explain.
 (b) Draw a graph between displacement from mean position and time for a body executing vibration in a medium.
- (ii) A person is standing at the sea shore. An observer on the ship which is anchored in between a vertical cliff and the person on the shore, fires a gun. The person on the shore hears two sounds, 2 seconds and 3 seconds after seeing the smoke of the fired gun. If the speed of sound in the air is 320 ms^{-1} then calculate:
 (a) the distance between the observer on the ship and the person on the shore.
 (b) the distance between the cliff and the observer on the ship.

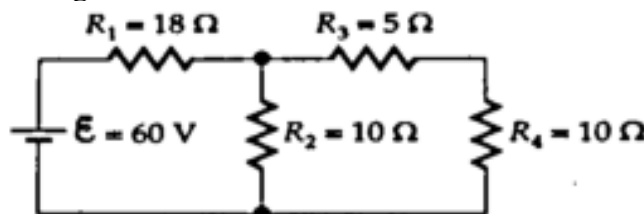


- (iii) A block and tackle pulley system has a velocity ratio of 5, with the effort applied in an inconvenient direction.
 (a) Draw a neat labelled diagram of the system indicating clearly the points of application and direction of load, effort and tension.
 (b) What will be its velocity ratio if the weight of the movable block is doubled?

Question 8

[3+3+4]

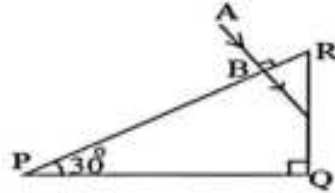
- (i) Emma loved playing the violin. One day, while practicing in her music room, she noticed something strange. When she played a particular note on her violin, a nearby glass on the shelf started to vibrate and eventually began to crack slowly. Curious, she experimented with different notes and realized that only one specific note caused the glass to shake strongly.
 (a) Name the phenomenon that causes the development of cracks.
 (b) Why did the glass develop cracks when Emma played that particular note?
- (ii) (a) A 1000 W electric kettle is used to boil water every day for 20 minutes. If the electricity cost is ₹5 per unit, calculate the monthly cost of using the kettle (30 days).
 (b) If the kettle's power rating is increased to 1500 W but the boiling time reduces to 15 minutes per day, which kettle will be more cost-effective?
- (iii) With reference to the following diagram, determine:
 (a) the voltage drop across the resistor R_1 .
 (b) the current flowing through R_2 .



Question 9

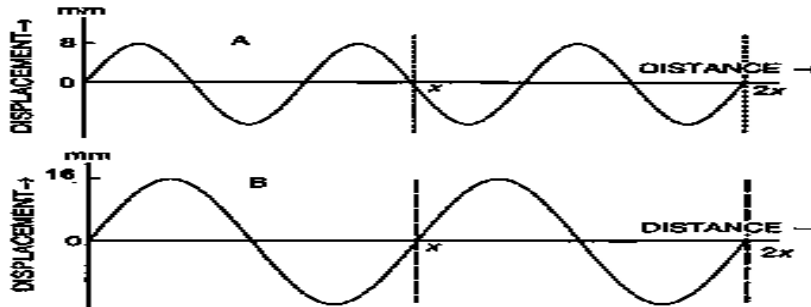
[3+3+4]

- (i) Complete the path of the ray AB through the glass prism PQR till it emerges out of the prism. Given the critical angle of the glass-air interface as 42° .



- (ii) The displacement-distance graphs for waves A and B are depicted below as they pass through air. What are the ratios between their:

(a) Wavelengths and (b) Loudness



- (iii) (a) A wire of length 100 cm under some tension produces a sound of frequency 256 Hz. What length of the same wire will produce a sound of frequency 1024 Hz, under the same tension?
 (b) The graph shows the displacement vs time relation of a disturbance travelling with velocity of 1500 ms^{-1} . Determine the wavelength of the disturbance.

